Experimental Report: Multilingual Word Embeddings for Indian Languages

Alok Sahni
PGDAI, SAIDE
Indian Institute of Technology Jodhpur
Jodhpur, India
g25ait2138@iitj.ac.in

***Abstract*—This paper trains and evaluates 300-dimensional word embeddings for seven Indian languages—Gujarati, Hindi, Marathi, Punjabi, Tamil, Telugu and Urdu—spanning six writing systems. One FastText Skip-gram model per language is trained from random initialisation on the IndicCorpV2-5k corpus only, with no pre-trained vectors. A layered out-of-vocabulary (OOV) resolver guarantees a valid vector for every benchmark word, so no word pair is discarded. Evaluated against the human similarity judgements of Akhtar et al., the models reach a macro-average Spearman correlation of $\rho = 0.2190$, ranging from 0.1571 (Marathi) to 0.3084 (Gujarati). A controlled ablation on Hindi and Tamil shows that the two languages respond best to opposite hyperparameter changes, and that vocabulary coverage is a weak predictor of embedding quality.**

***Index Terms*—word embeddings, FastText, Indian languages, subword models, out-of-vocabulary, word similarity**

I. Introduction

Distributional word embeddings map words to dense vectors whose geometry reflects co-occurrence statistics, so that words used in similar contexts receive similar vectors [2]. For Indian languages, two properties make this harder than for English: rich inflectional and agglutinative morphology, which produces many rare surface forms, and a variety of scripts whose combining marks must be handled correctly. Subword models such as FastText [1] address the first problem by representing a word as the sum of its character n-gram vectors, which also yields vectors for words never seen in training.

This work builds seven monolingual 300-dimensional FastText models from scratch on the IndicCorpV2-5k subset [3] and evaluates them intrinsically on the word-similarity datasets of Akhtar et al. [4].

A. Objectives and Scope

The study has three aims: (i) train embeddings from first principles under a fixed data budget, (ii) design an OOV strategy under which no evaluation pair is discarded, and (iii) measure how sensitive each language is to the context window, epoch budget and n-gram range. Two constraints shape the implementation: every benchmark word, including unseen inflections and malformed strings, must receive a vector, and the exported arrays must keep the original pair order with shape $(N, 2, 300)$. The benchmark is used for evaluation only; its human scores never enter training or model selection.

TABLE I
Learning Objectives and Where They Were Met

Objective	How it was met
Train distributional embeddings from first principles	Seven FastText Skip-gram models from random initialisation at exactly 300 dimensions, no pre-trained vectors (Sec. II-C)
Understand subword representation for rich morphology	Character n-gram composition as the main OOV mechanism, measured in the n-gram ablation and inflected-form test (Sec. II-D, V)
Design an OOV strategy instead of discarding cases	Four-layer resolver giving a valid vector for any input, including blank and malformed strings (Sec. II-D)
Apply correct intrinsic evaluation	Cosine similarity, Spearman correlation with p-values, macro-averaging and rank-based error analysis (Sec. II-F, VI)
Process multilingual Unicode text	NFC normalisation and Unicode-property tokenisation shared by training and evaluation across six scripts (Sec. II-B)
Handle imperfect real-world data	Mixed delimiters, BOMs, missing scores and a blank field handled while keeping every row in place (Sec. II-E)
Investigate hyperparameter sensitivity	Four-configuration ablation on two typologically distinct languages (Sec. V)
Interpret behaviour and state limitations	Diagnosis of best and worst predictions with failure causes; explicit limitations (Sec. VI, VII)

B. Learning Objectives

Table I pairs each learning objective of the assignment with the part of this work where it was exercised.

II. Methodology

A. Pipeline

Fig. 1 shows the end-to-end pipeline. Training text and benchmark words pass through the same normalisation and tokenisation path, which removes any train/evaluation mismatch.

B. Corpus, Normalisation and Tokenisation

The dataset `rony000013/indiccorpV2-5k`, loaded with the Hugging Face `datasets` library, contains 118,243 documents in one `train` split with the fields `language`, `code` and `text`. Rather than assuming a layout, the pipeline inspects the columns at runtime and resolves each record's language through an alias table (for example `hi`, `hin`, `hin_deva` and `hindi` all map to Hindi); in this run the language was read from the `code` column. This yields exactly 5,000 documents per language, and the pipeline stops if any

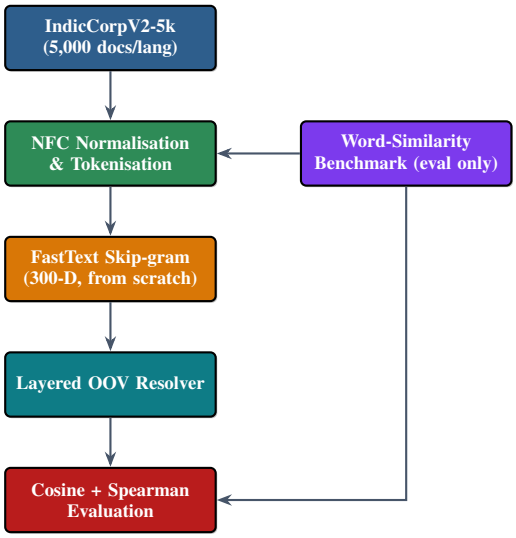


Fig. 1. Training and evaluation pipeline. Benchmark words share the training preprocessing path; human scores reach only the evaluation stage.

TABLE II
FASTTEXT CONFIGURATION (IDENTICAL FOR ALL LANGUAGES)

Parameter	Value	Parameter	Value
Architecture	Skip-gram	Negative samples	10
Vector size	300	Char n-grams	2–5
Window	5	Buckets	20,000
Min. count	2	Subsampling	10^{-4}
Epochs	15	Seed / workers	42 / 2

language cannot be identified, so no model is trained on text of the wrong language.

Text is normalised to Unicode NFC, case-folded and tokenised with the Unicode-property expression

```
[\p{L}\p{M}\p{N}]+(?:[-'][\p{L}\p{M}\p{N}]+)*
```

Because it matches letters, combining marks and digits by Unicode category rather than ASCII ranges, one code path serves all six scripts and keeps the vowel signs that carry meaning in Indic orthographies. For example, the Hindi sentence *acchā chātra vidyālay mẽ paṛhtā hai* ("a good student studies at school") yields six tokens, with the sentence-final danda removed. Sentences with fewer than two tokens are dropped from training.

C. Model and Hyperparameters

One independent Gensim [7] FastText model is trained per language from random initialisation with the configuration in Table II. FastText was chosen for its character n-gram composition, which is what makes principled OOV inference possible. Before the full run, a smoke test (100 documents, 2 epochs) verified imports, training, OOV inference and the $(N, 2, 300)$ export for every language. Its near-zero correlations (-0.14 to 0.08) are expected for such a tiny budget; smoke-test models are never saved or reported.

TABLE III
BENCHMARK ROWS EXCLUDED FROM THE CORRELATION ONLY

Language	Rows	Scored	Excl.	Reason
Marathi	253	104	149	Missing human score
Tamil	97	96	1	Blank word field
Other five	752	752	0	—

D. Out-of-Vocabulary Handling

Every input receives a finite, non-zero-norm vector through four layers: (i) in-vocabulary words use their learned vector; (ii) unseen words are composed from learned character n-grams; (iii) multi-token inputs are averaged; (iv) blank or degenerate inputs fall back to the frequency-weighted mean vocabulary vector

$$\mathbf{v}_{\text{fb}} = \frac{\sum_{w \in V} c_w \, \mathbf{v}_w}{\sum_{w \in V} c_w}, \tag{1}$$

where V is the model vocabulary, c_w the training count of word w and $\mathbf{v}_w$ its vector. Layer (iv) is the only one without word-specific information; it exists so that defective benchmark rows still produce a valid array entry rather than a zero vector. In this run it is reached only by one Tamil row with a blank word field, and no word pair is discarded at any stage.

E. Benchmark Integrity

The evaluation files contain real irregularities, which are handled rather than repaired. The Hindi file is comma-separated while the others are mostly tab-separated, so the delimiter is detected per line; files are decoded as `utf-8-sig` so byte-order marks are stripped; and missing or non-numeric scores become NaN instead of raising an error. No row is removed: all rows stay in the exported arrays in their original order, and a validity mask limits the Spearman computation to rows with two non-blank words and a finite score (Table III).

Evaluation-only guarantee: human similarity scores are used only to compute the reported correlations. They never enter training, never contribute gradients, and were not used for model selection or early stopping.

F. Evaluation Metric

For each pair (w_1, w_2), similarity is the cosine $s = \mathbf{v}_1^\top \mathbf{v}_2 / (\|\mathbf{v}_1\| \, \|\mathbf{v}_2\|)$. Per language, Spearman's rank correlation [5] between model scores s_i and human scores h_i is

$$\rho = \frac{\operatorname{cov}\big(R(h), R(s)\big)}{\sigma_{R(h)} \, \sigma_{R(s)}}, \tag{2}$$

where $R(\cdot)$ denotes ranks. The headline metric is the unweighted mean of ρ over the seven languages.

G. Implementation and Reproducibility

The pipeline is a single Colab notebook, seeded with 42 throughout. For every language it exports the artifacts in Table IV, and an inference interface (`embed_words`,

TABLE IV
EXPORTED ARTIFACTS PER LANGUAGE

Artifact	Content
`<Lang>.npy`	$(N, 2, 300)$ float32 embeddings in original pair order
`<Lang>_vocab.json`	Pair order and word-to-position index
`<Lang>_fasttext.model`	Trained Gensim FastText model
`<Lang>_pair_predictions.csv`	Human score, cosine and validity flag per pair
`tsne_<Lang>.png` + CSV	t-SNE plot and ID-to-word mapping

`embed_pairs`, `predict_similarity`) reloads the saved models, so new or private word pairs can be scored without retraining or access to benchmark scores.

Before packaging, a validator checks every language for (i) array shape $(N, 2, 300)$ and dtype float32, (ii) finite values and non-zero vector norms, (iii) exact agreement between the stored pair order and the benchmark file, and (iv) successful model reload with OOV inference on an unseen word. The final archive is created only if every check passes; in this run it holds 49 files.

H. Data Sources and External-Resource Disclosure

Training data: `rony000013/indiccorpV2-5k` only [3]. No pre-trained word embeddings, no external training corpora, and no use of the benchmark's human scores for training or supervised optimisation. *Evaluation data:* the word-similarity datasets of Akhtar et al. [4], used only to compute the reported correlations. *Tools:* Python, Gensim (FastText), NumPy, SciPy, Hugging Face `datasets`, `regex`, pandas, Matplotlib and scikit-learn (t-SNE). All links are listed under Project Resources at the end of this report.

III. T-SNE VISUALIZATION

Fig. 2 shows 100 word samples per language: the 100 most frequent vocabulary words of each model, projected from 300 to 2 dimensions with t-SNE [6] (perplexity 25, PCA initialisation, seed 42). The projections are diffuse rather than tightly clustered, consistent with the modest correlations and the small corpus. Dimensionality reduction is used only for this figure; all evaluation uses the full 300-dimensional vectors. Points are labelled with numeric IDs, mapped back to words in a CSV file per language, because the plotting environment lacks fonts for all six scripts.

IV. RESULTS

Cosine similarity was computed for all 1,102 benchmark pairs, and Spearman correlation (2) between these cosines and the human scores was computed on the 952 scored pairs of the public benchmark of Akhtar et al. [4]. Table V reports per-language results and Fig. 3 plots them. The macro-average is $\rho = 0.2190$. Gujarati performs best ($\rho = 0.3084$) and Marathi worst ($\rho = 0.1571$). Five of the seven correlations are significant at $p < 0.05$; Marathi ($p = 0.111$) and Tamil ($p = 0.072$) are not, and these are exactly the two languages with the fewest scored pairs (104 and 96), so their tests have the least statistical power.

TABLE V
PER-LANGUAGE RESULTS ON THE WORD-SIMILARITY BENCHMARK

Language	Pairs	Scored	ρ	p-value	Cov.
Gujarati	163	163	**0.3084**	0.00006	79.6%
Hindi	235	235	0.1901	0.0034	80.8%
Marathi	253	104	0.1571	0.1111	74.2%
Punjabi	143	143	0.2540	0.0022	82.5%
Tamil	97	96	0.1846	0.0718	75.0%
Telugu	111	111	0.2258	0.0172	67.5%
Urdu	100	100	0.2125	0.0337	86.3%
Macro / total	**1,102**	**952**	**0.2190**	—	**78.0%**

TABLE VI
VOCABULARY COVERAGE OF UNIQUE BENCHMARK WORDS

Language	Unique	In vocab	OOV	OOV rate
Gujarati	221	176	45	20.4%
Hindi	318	257	61	19.2%
Marathi	337	250	87	25.8%
Punjabi	206	170	36	17.5%
Tamil	136	102	34	25.0%
Telugu	160	108	52	32.5%
Urdu	146	126	20	13.7%
Total	**1,524**	**1,189**	**335**	**22.0%**

TABLE VII
OOV INFERENCE ON HINDI INPUTS (AFTER FULL TRAINING)

Pair (romanised)	Cosine	Case
vidyālay / *skūl*	0.6843	Synonyms, both in vocabulary
vidyālayõ / *vidyālay*	0.8966	Inflected plural, OOV in training
malformed string / *vidyālay*	0.6239	Garbled input, resolved via subwords

A. Coverage and OOV Behaviour

Table VI gives vocabulary coverage over the unique words in each benchmark. Coverage of unique benchmark words ranges from 67.5% (Telugu) to 86.3% (Urdu), yet it barely predicts quality: Urdu has the highest coverage but ranks fourth on ρ, while Telugu has the lowest coverage and ranks third. Across languages, the rank correlation between coverage and ρ is only 0.29. Table VII shows the resolver on Hindi inputs of increasing difficulty. The plural *vidyālayõ* ("schools"), absent from training, scores 0.897 against its base form—higher than the in-vocabulary synonym pair—because both forms share most of their character n-grams. The malformed third string still receives a finite vector, confirming that no input can break the inference path.

V. EXPERIMENTS

Four configurations were trained on Hindi (Indo-Aryan, moderately inflecting) and Tamil (Dravidian, agglutinative), each from scratch at 300 dimensions on IndicCorpV2 only (Table VIII, Fig. 4). Benchmark scores were used only to read off the resulting correlation, never as an optimisation target.

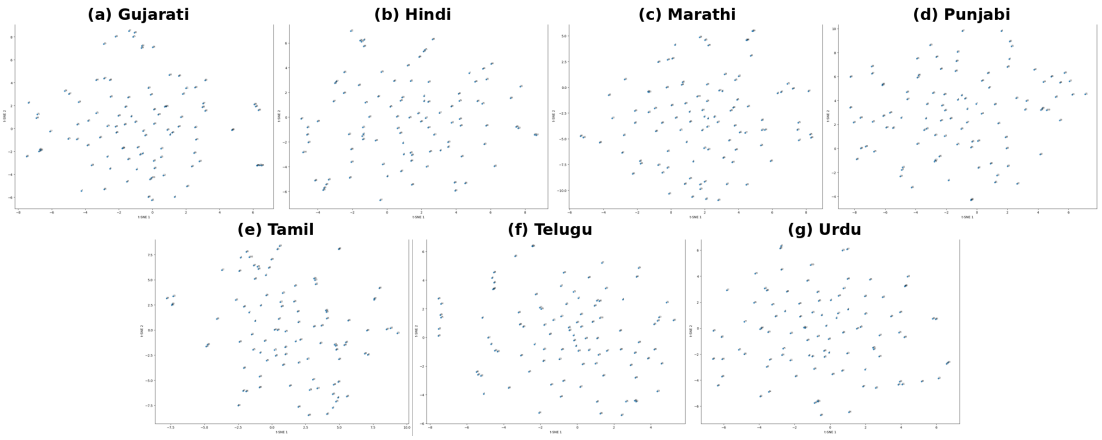


Fig. 2. t-SNE projections of the 100 most frequent words per language. Points carry numeric IDs mapped to words in per-language CSV files.

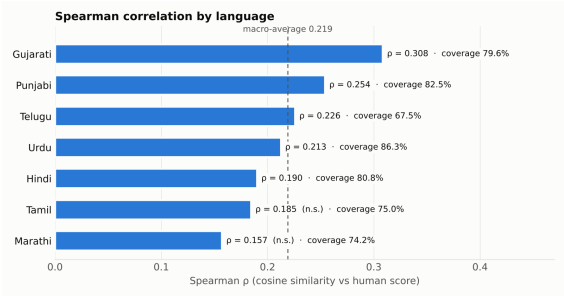


Fig. 3. Spearman correlation by language with vocabulary coverage; n.s. marks correlations not significant at $p < 0.05$.

TABLE VIII
ABLATION RESULTS (SPEARMAN ρ)

Run	Win.	Ep.	n-grams	Hindi	Tamil
E1 baseline	5	15	2–5	0.1982	0.1988
E2 wider window	10	15	2–5	0.2654	0.1847
E3 longer training	5	30	2–5	**0.3181**	0.1401
E4 larger n-grams	5	15	3–6	0.1564	**0.2891**

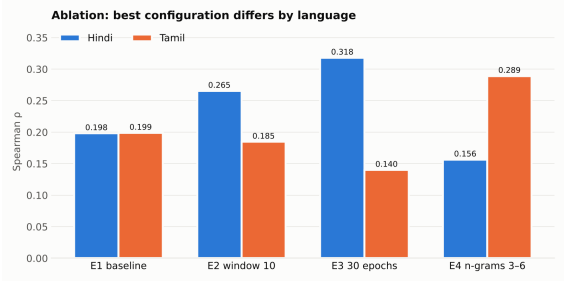


Fig. 4. Ablation on Hindi and Tamil. The best configuration differs by language.

The two languages respond to opposite interventions. Doubling the epochs lifts Hindi from 0.1982 to 0.3181 (+61%) but lowers Tamil from 0.1988 to 0.1401 (−30%), the lowest score in the study. A wider window also helps Hindi (+34%) while slightly hurting Tamil (−7%). Conversely, 3–6 character n-grams lift Tamil to 0.2891 (+45%) and give Hindi its worst score, 0.1564. Tamil packs meaning into longer character sequences, so 3–6 grams capture morpheme-sized units that 2–5 grams fragment; Hindi's shorter affixes are already served by 2–5 grams, and its model was still improving at 15 epochs. With the short 2–5 range, extra epochs on Tamil appear to reinforce short, uninformative fragments rather than add signal. A single hyperparameter set applied uniformly across Indian languages therefore leaves substantial performance unclaimed. Because training uses two worker threads, identical configurations differ by about 0.01 between runs (Hindi: 0.1982 in E1 versus 0.1901 in the main run), well below the observed effects of 0.05–0.12.

VI. ANALYSIS

Because human scores (1–10) and cosines ($[-1, 1]$) differ in scale, pairs are compared by absolute rank error. Table IX lists ten word-pair examples: for each language, the pair the model ranked closest to the annotators (correctly predicted) and the pair it ranked furthest from them (incorrectly predicted).

A. Correctly Predicted Similarities

Four of the five closest agreements are identity pairs—the same word compared with itself, scored 10 by annotators and 1.000 by cosine. They confirm the pipeline is wired correctly but say little about semantics. The informative success is Hindi *bāgh/ciṛiyāghar* ("tiger"/"zoo", human 6.80, cosine 0.502): two different words that co-occur in news text about wildlife and zoos, which the model places almost exactly at the rank the annotators chose.

B. Incorrectly Predicted Similarities

The failures share a structure. Hindi *paise/mudrā* ("money"/"currency", human 8.60, cosine 0.271) and Punjabi *paisā/nakad* ("money"/"cash") are near-synonyms from different registers—one colloquial, one formal or Sanskritic. In a 5,000-document corpus these registers rarely share contexts, so the model separates them.

TABLE IX
CLOSEST AND FURTHEST RANK AGREEMENT (ROMANISED)

Lang.	Pair (gloss)	Hum.	Cos.	Err.
Closest agreement				
Guj.	*vāgh* / *vāgh* (tiger)	10.00	1.000	0.0
Hin.	*bāgh* / *cid̤iyāghar* (tiger / zoo)	6.80	0.502	0.5
Mar.	*vāgh* / *vāgh* (tiger)	10.00	1.000	0.0
Pun.	*ṭāīgar* / *ṭāīgar* (tiger)	10.00	1.000	0.0
Tam.	*paṇam* / *paṇam* (money)	10.00	1.000	0.5
Largest disagreement				
Guj.	*so* / *ṭakā* (hundred / percent)	9.00	0.053	138.5
Hin.	*paise* / *mudrā* (money / currency)	8.60	0.271	203.5
Mar.	*ṭeliphon* / *saṃvād* (telephone / communication)	9.00	0.225	91.0
Pun.	*paisā* / *nakad* (money / cash)	9.00	0.325	116.5
Tam.	*vaṅki* / *paṇam* (bank / money)	8.75	0.053	84.0

Gujarati *so*/*ṭakā* ("hundred"/"percent", cosine 0.053) is a fixed collocation rather than a shared context. Tamil *vaṅki*/*paṇam* ("bank"/"money") and Marathi *ṭeliphon*/*saṃvād* ("telephone"/"communication") rest on world knowledge that a small corpus does not supply consistently; the Hindi tiger/zoo pair shows the model can learn such links when co-occurrence evidence exists. These are the expected failure modes of a distributional model trained on little text, not implementation defects.

VII. DISCUSSION

A. Design Choices

FastText was preferred to Word2Vec [2] and GloVe because, with 22.0% of benchmark vocabulary unseen on average, the no-discard requirement effectively needs a subword model. Seven monolingual models were trained rather than one multilingual model because the benchmark evaluates each language in isolation, so there is no cross-lingual signal to exploit, and separate models avoid vocabulary interference between scripts. Preprocessing is shared between training and evaluation to eliminate normalisation drift, a common and silent cause of inflated OOV rates. Every benchmark row is kept in the exported arrays so the $(N, 2, 300)$ shape and pair order are preserved exactly.

B. Limitations

Corpus size dominates: 5,000 documents per language is a thin slice, and Hindi was still improving at 30 epochs, so the reported correlations are a floor set by data rather than a ceiling set by the method. The benchmarks hold 96–235 scored pairs, so individual correlations carry wide confidence intervals. Static embeddings assign one vector per word form, so polysemy cannot be resolved by construction. Multi-threaded training makes repeated runs differ by about 0.01 in ρ, and each ablation configuration was run once. Finally, the ablation covers only two languages and four configurations: enough to show that per-language tuning matters, not enough to find optimal settings for the other five.

C. Future Work

In order of expected value: scale the training corpus and re-measure the coverage–correlation relationship; run a per-language hyperparameter search with several seeds, now that the optima are known to diverge; add Word2Vec and GloVe baselines under the same evaluation harness to quantify the subword contribution; and validate the embeddings on a downstream task, since intrinsic similarity is only a weak proxy for usefulness.

D. Conclusion

Seven monolingual 300-dimensional FastText models trained from scratch on IndicCorpV2 reach a macro-average Spearman correlation of 0.2190 against human similarity judgements, with all 1,102 benchmark pairs represented in the exported arrays. Three findings stand out: subword composition makes the no-discard requirement satisfiable, as the unseen plural *vidyālayõ* still scores 0.897 against its base form; hyperparameters do not transfer across Indian languages, as Hindi gained 61% from a longer epoch budget that cost Tamil 30%, while Tamil gained 45% from longer n-grams that gave Hindi its worst result; and vocabulary coverage is not a proxy for embedding quality, as the language with the highest coverage ranked only fourth. The correlations are modest because the models are corpus-limited rather than method-limited.

VIII. REFERENCES

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PROJECT RESOURCES

Resource	Link
Training corpus — IndicCorpV2-5k	https://huggingface.co/datasets/rony000013/indiccorpV2-5k
Evaluation benchmark — Akhtar et al. (2017)	https://github.com/syedsarfarazakhtar/Word-Similarity-Datasets-for-Indian-Languages
Project source code and artifacts	https://github.com/g25ait2138-collab/word-embeddings-nlp